

Dong Heon Han

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Robotics Department, University of Michigan, Ann Arbor, MI, USA

SUMMARY

1st year PhD student developing large-scale (1024-channel) electromagnetic platforms for microrobot swarms, enabling programmable collective behaviors and spatial field control.

EDUCATION

- **University of Michigan** 2029
PhD in Robotics Ann Arbor, MI
- **University of Michigan** 2025
MS in Mechanical Engineering Ann Arbor, MI
- **Georgia Institute of Technology** 2021
BS in Mechanical Engineering Atlanta, GA

WORK EXPERIENCE

- **University of Michigan | Robotics Department** June 2025 - Present
Research Assistant | Advisor: Dr. Steven Ceron Ann Arbor, MI
- **University of Michigan | Mechanical Engineering Department** Aug 2023 - May 2025
Research Assistant | Advisor: Dr. Daniel Bruder Ann Arbor, MI
- **Republic of Korea Army** Jan 2022 - July 2023
Signal Specialist Korea
- **Seoul National University | Biosystems Engineering Department** Aug 2021 - Dec 2021
Research Assistant | Advisor: Dr. Young-Jun Park Seoul, Korea
- **George W. Woodruff School of Mechanical Engineering** Aug 2019 - Aug 2021
Research Assistant | Advisor: Dr. Ye Zhao Atlanta, GA
- **Korean Institute of Machinery and Materials** Jun 2018 - Aug 2018
Research Intern Daejeon, Korea

PATENTS AND PUBLICATIONS

C=CONFERENCE, J=JOURNAL, PAT=PATENT, S=IN SUBMISSION, P=PREPARATION, T=THESIS

- [C.1] **D.H. Han**, X. Xu, Y. Chen, Y. Zhou, X. Zhang, D. Bruder, X. Huang. "Towards Whole-Body Proprioceptive Morphing: A Soft Gripper for Cross-Scale Grasping". 2026 IEEE International Conference on Soft Robotics (RoboSoft).
- [C.2] R. Zuo, M. Mehta, **D.H. Han**, D. Bruder. "Embedded Valves for Distributed Control of Soft Pneumatic Actuators". 2024 IEEE International Conference on Intelligent Robots and Systems (IROS)
- [C.3] **D.H. Han**, S.J. Byeon, K.D. Kim, G.H. Han, M.H. Cha, Y.J. Park. "Development of Path Tracking Control Algorithm for Tractor Autonomous Driving". 2021 Korean Society for Agricultural Machinery Conference
- [Pat.1] **Blowers With Variable Nozzles**. US 11668311 B2. Issued June 6, 2023.
- [S.1] **D.H. Han**, M. Mehta, R. Zuo, Z. Wanger, and D. Bruder. "An Enhanced Proprioceptive Method for Soft Robots Integrating Bend Sensor and IMU"
- [S.2] R. Zuo, **D.H. Han**, R Li, S. Jamal, D. Bruder. "UMArm: Untethered, Modular, Wearable, Soft Pneumatic Arm with 12 DoF"
- [S.3] **D.H. Han**, R. Zuo, C. Kim, S. Jamal, D. Bruder. "BendSense: A Modular Off-the-Shelf Bend Sensor Framework for Proprioception in Soft Robots"

HONORS AND AWARDS

- **VIP Innovation Competition, 1st Place in Hardware, Devices & Robotics Track** Apr 2021
Georgia Institute of Technology
- **President's Undergraduate Research Award** Oct 2020
Georgia Institute of Technology
- **Georgia Korean American Grocers Association Scholarship Award** Dec 2016
KAGRO

CORE SKILLS

- **Robotics:** System-level robotic design and integration, hardware/firmware/software co-design, ROS2
- **Programming:** C, C++, Python, MATLAB
- **Design and Fabrication:** Embedded systems, CAD, 3D printing, silicone casting, rapid prototyping